

Merchant Onboarding, Payment Processing & Risk Management Platform

Business Analyst Case Study for a Digital Merchant Acquiring Platform

Executive Overview

A digital payment provider is experiencing operational inefficiencies due to:

- Manual merchant onboarding and KYC verification
- Delayed payment terminal activation
- Increasing fraudulent transaction attempts
- Slow settlement processing
- Limited operational visibility into merchant performance

The objective is to design an end-to-end platform that automates merchant onboarding, strengthens fraud controls, streamlines settlement processing, and provides actionable business insights through performance reporting dashboards.

Business Capabilities

- Merchant Registration
- KYC Verification
- Terminal Provisioning
- Payment Processing
- Fraud Detection
- Settlement Engine
- Merchant Operations Dashboard
- Business Insights & Reporting

Project Deliverables

Asset 1 (PPT)

Executive Case Study

Asset 2 (Word)

GitHub

- SQL Schema
- Sample Data
- SQL Queries
- Tableau Dashboard
- README

SECTION 1 — Business Context

Digital payment providers are experiencing rapid growth in merchant acquisitions across retail, e-commerce, SaaS, healthcare, and service-based businesses. As merchant volumes increase, onboarding operations become more complex due to regulatory compliance requirements, fraud prevention measures, and settlement obligations.

In many organizations, merchant onboarding still relies on multiple manual verification steps, including business document validation, tax identifier verification, risk assessment, and payment terminal provisioning. These fragmented processes increase operational effort, delay merchant activation, and impact the overall merchant experience.

Additionally, payment providers must continuously monitor transaction activity to identify potentially fraudulent payments before settlement. Delayed fraud detection can expose the organization to financial losses, chargebacks, and regulatory risks. Settlement operations also require accurate reconciliation to ensure only eligible transactions are included in merchant payouts.

To address these challenges, the organization plans to implement an integrated Merchant Onboarding, Payment Processing & Risk Management Platform that automates merchant verification, streamlines payment processing, strengthens fraud controls, and improves executive visibility through real-time analytics and reporting.

Business Drivers

- Reduce merchant onboarding turnaround time through automated KYC verification.
- Improve merchant experience by enabling faster payment terminal activation.
- Strengthen fraud detection before transaction settlement.

- Automate settlement processing and payout reconciliation.
- Improve performance visibility using centralized dashboards and performance reporting.
- Support increasing merchant volumes through an integrated merchant management workflow.

Expected Business Outcomes

- Faster merchant onboarding and activation.
- Reduced manual effort through workflow automation.
- Improved fraud detection and settlement accuracy.
- Better operational decision-making through centralized reporting

SECTION 2 — Problem Statement

Problem Statement

The existing merchant onboarding process relies heavily on manual verification, disconnected operational workflows, and delayed compliance checks. Merchant registration details are validated through multiple independent systems, resulting in longer onboarding cycles, inconsistent decision-making, and increased operational effort.

The absence of real-time risk assessment during payment processing increases exposure to fraudulent transactions, chargebacks, and financial losses. Additionally, settlement processing depends on manual reconciliation activities, delaying merchant payouts and reducing operational efficiency.

Limited visibility into merchant onboarding status, transaction health, fraud cases, and settlement performance further restricts the organization's ability to make timely operational decisions and proactively manage risks.

As the business continues to onboard more merchants and process higher transaction volumes, the existing operating model becomes increasingly difficult to scale while maintaining compliance, security, and service quality.

Key Business Challenges

- Manual KYC verification increases merchant onboarding turnaround time.
- Fragmented verification processes create operational inefficiencies.
- Delayed terminal activation impacts merchant experience.
- Limited real-time fraud detection increases financial risk.

- Manual settlement reconciliation delays merchant payouts.
- Lack of centralized reporting reduces operational reporting.
- Existing processes do not scale efficiently with growing merchant volumes.

Business Impact

- Increased merchant onboarding delays.
- Higher operational costs due to manual effort.
- Greater exposure to fraud and chargebacks.
- Delayed settlement cycles affecting merchant satisfaction.
- Limited data-driven decision-making.
- Reduced scalability for future business growth.

SECTION 3 — Project Objectives

Project Objectives

The primary objective of this initiative is to design a proposed centralized Merchant Onboarding, Payment Processing & Risk Management Platform that automates merchant lifecycle management, strengthens regulatory compliance, and improves operational efficiency across payment processing and settlement activities.

The solution aims to streamline merchant onboarding through automated KYC verification, enable secure payment processing with real-time fraud monitoring, accelerate settlement operations, and provide business stakeholders with actionable operational insights through centralized reporting and analytics dashboard. The platform is designed to improve merchant experience while ensuring regulatory compliance, operational scalability, and secure payment processing.

Business Objectives

- Reduce merchant onboarding turnaround time through automated KYC verification.
- Enable faster payment terminal activation for eligible merchants.
- Improve fraud detection by introducing real-time transaction risk assessment.
- Automate settlement processing to ensure timely and accurate merchant payouts.
- Improve operational efficiency by reducing manual verification and reconciliation activities.

- Provide business intelligence dashboards to monitor merchant onboarding, transaction performance, fraud cases, and settlement status.
- **Build an integrated solution capable of supporting increasing merchant and transaction volumes** while maintaining regulatory compliance.

Success Criteria

- Reduced merchant onboarding turnaround time.
- Improved merchant activation rate.
- Increased fraud detection before settlement processing.
- Reduced manual operational effort across onboarding and settlements.
- Improved settlement accuracy and payout timeliness.
- Enhanced performance visibility through centralized dashboards and reporting.
- Improved scalability to support increasing merchant volumes.

SECTION 4 — Project Scope

Project Scope

This project focuses on designing and documenting a proposed integrated Merchant Onboarding, Payment Processing & Risk Management Platform that automates merchant lifecycle processes, strengthens compliance controls, and improves operational efficiency across payment operations.

The solution covers merchant onboarding, KYC verification, payment processing, fraud monitoring, settlement management, and operational reporting to support secure and scalable merchant acquiring operations.

In Scope

- Merchant registration and profile management.
- Automated KYC verification and compliance workflow.
- Payment terminal provisioning and activation.
- Payment transaction processing.
- Real-time fraud detection and transaction hold management.
- Settlement processing and payout management.
- Merchant operational dashboards and reporting.

- REST API integrations with external verification and payment services.
- Audit logging and operational tracking.
- SQL-based analytics and business intelligence reporting.

Out of Scope

- Customer-facing payment gateway UI.
- Mobile applications for merchants or customers.
- Banking core system implementation.
- Third-party accounting and ERP integrations.
- Multi-country regulatory compliance and taxation.
- AI/ML fraud model development.
- Infrastructure deployment, DevOps, and cloud provisioning.
- Customer mobile and web application development.

Key Deliverables

- BRD
- Functional Requirements Specification
- Business Rules
- AS-IS / TO-BE
- System Architecture
- API Specifications
- User Stories
- Wireframes
- ERD
- SQL Schema
- SQL Analytics
- Tableau Dashboard
- Risk Assessment & Success Metrics

SECTION 5 — Stakeholder Matrix

Stakeholder	Role	Responsibilities	Key Expectations
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Product Manager / Product Owner	Business Sponsor	Defines business vision, priorities, and product goals.	Faster merchant onboarding, improved merchant experience, and business growth.
Senior Business Analyst	Requirements Lead	Elicits, analyzes, documents, and validates business and functional requirements while coordinating across stakeholders.	Well-defined requirements, complete documentation, and successful project delivery.
Engineering Lead	Technical Owner	Designs and develops the platform, APIs, integrations, and backend services.	Clear functional requirements, API specifications, and business rules.
Risk & Compliance Team	Compliance Owner	Defines KYC policies, fraud controls, and regulatory requirements.	Regulatory compliance, secure onboarding, and fraud prevention.
Merchant Operations Team	Operations	Manages merchant onboarding, exception handling, and settlement support.	Reduced manual effort and streamlined operational workflows.
Finance & Settlement Team	Financial Operations	Oversees settlement processing, payout reconciliation, and financial reporting.	Accurate settlements, timely payouts, and reconciliation visibility.
QA Team	Quality Assurance	Validates functional and non-functional requirements through testing.	Complete acceptance criteria and defect-free releases.
External Verification Services	Third-Party Integration	Provides KYC, tax verification, and regulatory validation through APIs.	Reliable and secure API communication.

Stakeholder Communication Plan

Stakeholder	Communication Type	Frequency
Product Manager / Product Owner	Sprint Review & Business Updates	Weekly

Engineering Lead	Requirement Clarification & Design Discussion	Weekly
Risk & Compliance Team	Compliance Review	As Required
Merchant Operations Team	Process Review & UAT Feedback	Weekly
Finance & Settlement Team	Settlement Review	Bi-weekly
QA Team	Sprint Planning, Testing & Defect Review	Every Sprint

SECTION 6 — Success Metrics

The success of the Merchant Onboarding, Payment Processing & Risk Management Platform will be measured using operational, compliance, financial, and customer experience KPIs. These metrics will help stakeholders evaluate platform performance, operational efficiency, and business impact after implementation.

KPI	Description	Business Objective
Merchant Onboarding Turnaround Time	Average time required to complete merchant onboarding from registration to activation.	Reduce onboarding delays and improve merchant experience.
Merchant Activation Rate	Percentage of registered merchants successfully activated after KYC verification.	Improve onboarding efficiency.
KYC Approval Rate	Percentage of merchant applications approved after compliance verification.	Strengthen regulatory compliance.
Fraud Detection Rate	Percentage of high-risk transactions identified before settlement.	Minimize financial risk and fraud exposure.
Settlement Processing Time	Average time required to complete merchant settlements.	Accelerate merchant payouts.
Settlement Accuracy	Percentage of settlements processed without reconciliation issues.	Improve financial accuracy.
Operational Automation Rate	Percentage of onboarding and settlement activities completed without manual intervention.	Reduce operational effort.

Transaction Success Rate	Percentage of successfully processed payment transactions.	Improve payment reliability.
Dashboard Adoption	Usage of operational dashboards by business and operations teams.	Improve data-driven decision making.

Expected Business Benefits

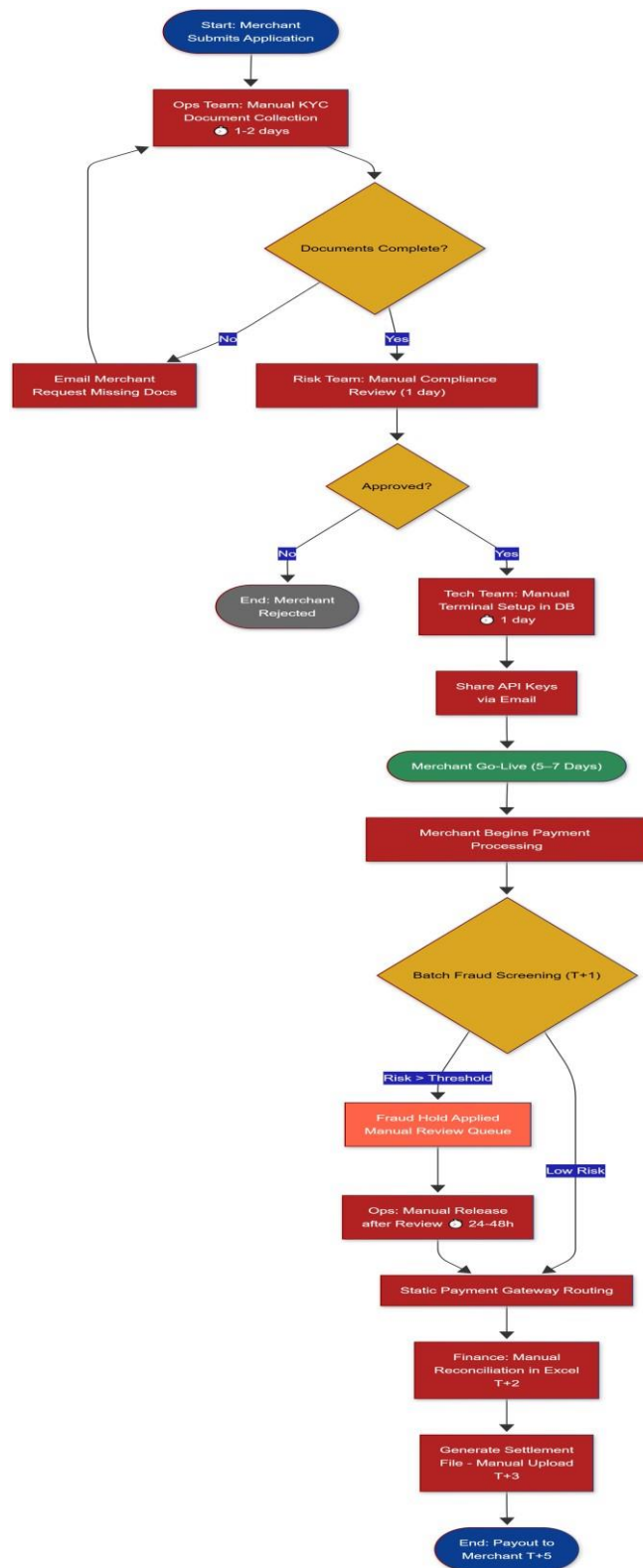
- Reduced merchant onboarding turnaround time.
- Faster merchant activation and improved onboarding experience.
- Improved compliance through automated KYC verification.
- Reduced fraud-related financial exposure.
- Faster and more accurate settlement processing.
- Increased operational efficiency through process automation.
- Improved business visibility using centralized operational dashboards.
- Modular platform architecture designed to support additional merchant volumes.

SECTION 7 — AS-IS Process

The current merchant onboarding and payment processing workflow relies on multiple manual activities performed across different operational teams. Merchant verification, payment terminal activation, fraud monitoring, and settlement processing are executed through disconnected systems, resulting in delays, operational inefficiencies, and limited process visibility.

The absence of an integrated workflow increases dependency on manual validations, slows merchant activation, and makes fraud detection and settlement reconciliation reactive rather than proactive.

Current Process Flow



Current Process Challenges

Process Area	Current Challenge	Business Impact
Merchant Registration	Manual document verification	Increased onboarding time
KYC Verification	Multiple systems and manual checks	Operational delays and inconsistent decisions
Terminal Provisioning	Manual activation process	Delayed merchant go-live
Fraud Monitoring	Fraud identified after transaction processing	Higher financial exposure
Settlement Processing	Manual reconciliation	Delayed merchant payouts
Reporting	Limited operational visibility	Slow business decision-making

Key Pain Points

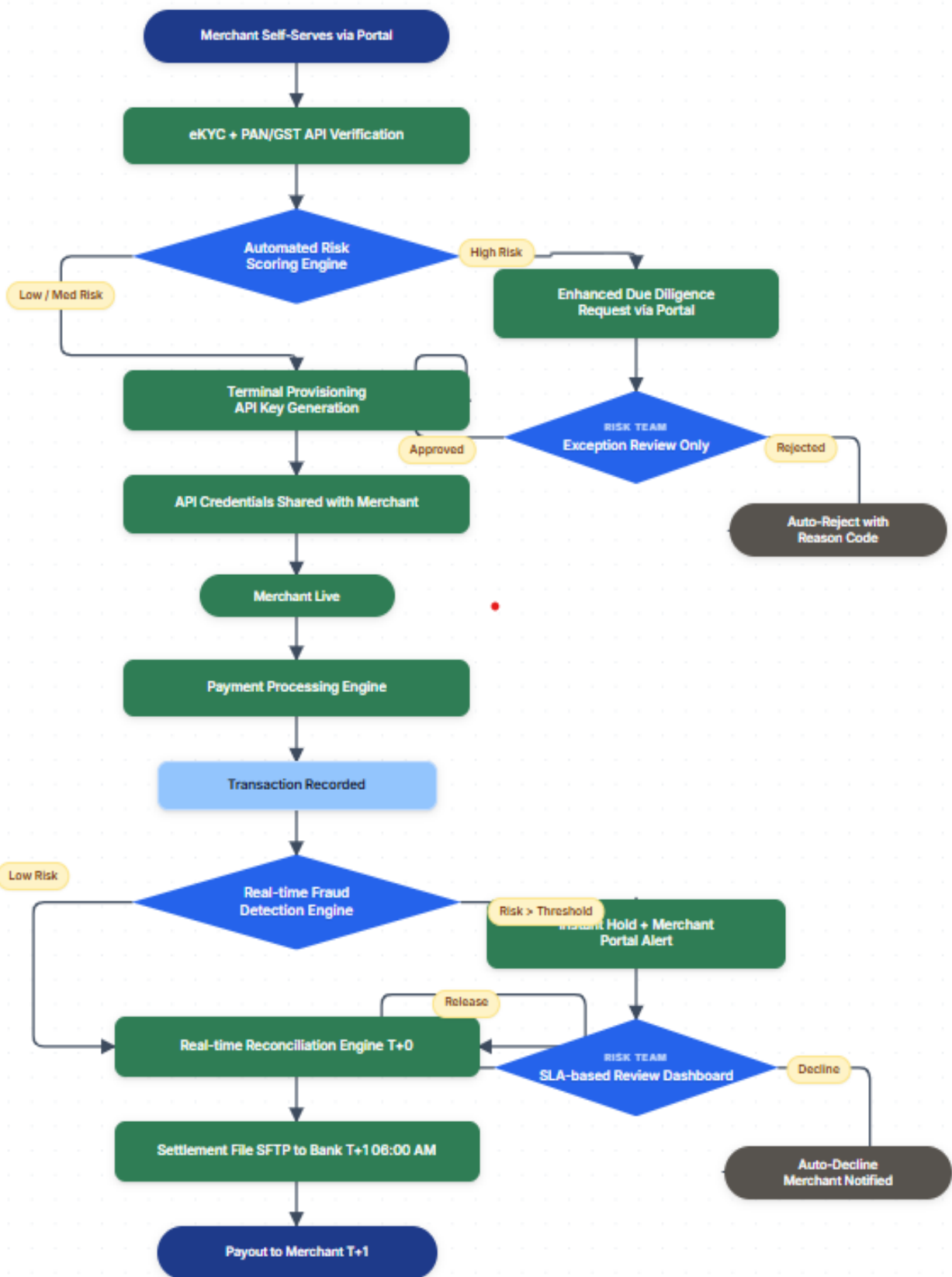
- Manual KYC verification increases onboarding turnaround time.
- Multiple handoffs between operational teams delay merchant activation.
- Fraud detection occurs after transaction execution instead of proactively.
- Settlement processing requires manual reconciliation activities.
- Operational teams lack centralized visibility into merchant and transaction status.
- Existing workflows are difficult to scale as merchant volumes increase.

SECTION 8 — TO-BE Process

The proposed solution introduces an integrated Merchant Onboarding, Payment Processing & Risk Management Platform that automates merchant verification, payment processing, fraud monitoring, and settlement operations through a centralized workflow.

The platform minimizes manual intervention by integrating external KYC verification services, real-time fraud assessment, automated terminal provisioning, and settlement processing into a single operational ecosystem. This enables faster merchant activation, improved compliance, enhanced operational efficiency, and greater business visibility.

Future Process Flow



Process Improvements

Process Area	Current State	Future State
Merchant Onboarding	Manual verification	Automated onboarding workflow
KYC Verification	Manual compliance checks	API-based real-time verification
Terminal Activation	Manual provisioning	Automatic activation after approval
Fraud Monitoring	Post-transaction review	Real-time risk assessment
Settlement	Manual reconciliation	Automated settlement processing
Reporting	Multiple disconnected reports	Centralized operational dashboard

Expected Operational Benefits

- Reduced merchant onboarding turnaround time.
- Faster merchant activation through automated verification.
- Improved fraud detection before settlement execution.
- Reduced manual effort across onboarding and settlement operations.
- Faster merchant payouts with automated settlement workflows.
- Improved compliance through standardized verification processes.
- Enhanced executive visibility through centralized dashboards.
- Improved scalability to support future business growth.

SECTION 9 — Gap Analysis

The assessment of the current (AS-IS) and proposed (TO-BE) processes highlights several operational and technology gaps that limit business efficiency, regulatory compliance, and scalability. The proposed platform addresses these gaps by introducing automated business processes, system integrations, real-time monitoring, and centralized performance visibility.

Gap Analysis Matrix

Business Area	AS-IS (Current State)	TO-BE (Future State)	Business Benefit
Merchant Onboarding	Manual registration and document verification	Automated onboarding workflow with centralized tracking	Reduced onboarding turnaround time

KYC Verification	Manual verification across multiple systems	API-driven real-time KYC validation	Faster approvals and improved compliance
Terminal Provisioning	Manual terminal activation	Automatic terminal activation after KYC approval	Faster merchant go-live
Payment Processing	Limited process visibility	Centralized transaction processing and monitoring	Improved operational control
Fraud Management	Fraud detected after transaction processing	Real-time fraud assessment before settlement	Reduced financial risk
Settlement Processing	Manual reconciliation and payout processing	Automated settlement workflow	Faster and more accurate merchant payouts
Reporting & Analytics	Disconnected reports from multiple systems	Centralized operational dashboards	Improved decision-making and visibility
Scalability	High dependency on manual operations	Automated, scalable workflows	Supports future merchant and transaction growth

Key Process Improvements

- Automate merchant onboarding to reduce manual effort.
- Enable API-based KYC verification for faster compliance checks.
- Provision payment terminals automatically after successful merchant approval.
- Introduce real-time fraud monitoring before settlement processing.
- Automate settlement reconciliation and merchant payout workflows.
- Centralize operational reporting through business intelligence dashboards.
- Improve scalability by reducing dependency on manual operations.

Business Value Delivered

- Improved merchant experience through faster onboarding.
- Reduced operational costs by minimizing manual intervention.
- Strengthened regulatory compliance through standardized verification.
- Reduced fraud exposure and financial losses.
- Improved settlement accuracy and payout efficiency.

- Enhanced operational transparency through centralized reporting.
- Modular platform architecture designed to support additional capabilities with limited impact on existing workflows.

SECTION 10 — Functional Requirements

The proposed platform shall provide end-to-end capabilities to automate merchant onboarding, payment processing, fraud monitoring, settlement operations, and business reporting. The functional requirements are grouped into logical business modules to support efficient development, testing, and implementation.

FR-01: Merchant Management

Requirement ID	Functional Requirement
FR-01.1	The system shall allow merchants to register using business and contact information.
FR-01.2	The system shall create a unique Merchant ID for every successful registration.
FR-01.3	The system shall maintain merchant profile details and onboarding status.
FR-01.4	The system shall allow authorized users to update merchant information.
FR-01.5	The system shall maintain a complete audit trail of profile updates.

FR-02: KYC & Compliance

Requirement ID	Functional Requirement
FR-02.1	The system shall validate merchant KYC details through external verification APIs.
FR-02.2	The system shall assign KYC status as Pending Review, Approved, or Rejected.
FR-02.3	The system shall prevent merchant activation until KYC is approved.
FR-02.4	The system shall record all KYC verification responses for audit purposes.
FR-02.5	The system shall notify operations teams when manual review is required.

FR-03: Terminal Management

Requirement ID	Functional Requirement
FR-03.1	The system shall provision a payment terminal after successful KYC approval.
FR-03.2	The system shall generate unique terminal credentials for each merchant.
FR-03.3	The system shall allow terminal activation or deactivation by authorized users.
FR-03.4	The system shall maintain terminal status and activation history.

FR-04: Payment Processing

Requirement ID	Functional Requirement
FR-04.1	The system shall process payment transactions initiated through active merchant terminals.
FR-04.2	The system shall validate transaction requests before processing.
FR-04.3	The system shall record transaction status as Success, Failed, or Suspicious Hold.
FR-04.4	The system shall generate unique transaction identifiers for every payment.
FR-04.5	The system shall maintain transaction history for reporting and reconciliation.

FR-05: Fraud Management

Requirement ID	Functional Requirement
FR-05.1	The system shall evaluate each transaction using a fraud risk score.
FR-05.2	The system shall place high-risk transactions under review before settlement.
FR-05.3	The system shall allow authorized analysts to review and release held transactions.
FR-05.4	The system shall maintain complete fraud investigation history.

FR-06: Settlement Management

Requirement ID	Functional Requirement
FR-06.1	The system shall calculate merchant settlement amounts based on successful transactions.
FR-06.2	The system shall exclude transactions under fraud review from settlement.
FR-06.3	The system shall generate settlement records for each payout cycle.
FR-06.4	The system shall maintain settlement status and payout reference details.
FR-06.5	The system shall maintain transaction-level settlement details for each settlement record to support reconciliation and auditability.

FR-07: Reporting & Analytics

Requirement ID	Functional Requirement
FR-07.1	The system shall provide dashboards for merchant onboarding performance.
FR-07.2	The system shall display transaction and settlement metrics.
FR-07.3	The system shall provide fraud monitoring reports.
FR-07.4	The system shall support SQL-based analytical reporting for business insights.

FR-08: Notifications & Audit

Requirement ID	Functional Requirement
FR-08.1	The system shall notify merchants of onboarding status changes.
FR-08.2	The system shall notify operations teams of pending compliance reviews.
FR-08.3	The system shall maintain audit logs for all critical business events.
FR-08.4	The system shall capture user activity for compliance and traceability.

Status / State Transitions

Entity	Status Flow
Merchant	Pending Review → Approved / Rejected
Terminal	Provisioning → Active / Failed → Deactivated
Transaction	Initiated → Success / Failed → Fraud Assessment → Low Risk → Settled / Suspicious Hold → Released → Settled / Rejected → Not Settled

SECTION 11 — Non-Functional Requirements

Non-Functional Requirements

The platform shall meet defined quality, performance, security, reliability, and scalability standards to ensure stable and secure operations while supporting future business growth.

NFR-01: Performance

Requirement ID	Non-Functional Requirement
NFR-01.1	The system shall support the expected level of concurrent merchant onboarding requests within the performance threshold agreed during non-functional requirements validation.
NFR-01.2	The system shall process payment transactions within the response-time threshold agreed during non-functional requirements validation.
NFR-01.3	The platform shall generate operational dashboards and reports within the response-time threshold agreed during non-functional requirements validation.

NFR-02: Availability

Requirement ID	Non-Functional Requirement
NFR-02.1	The platform shall be available during agreed business operating hours with minimal planned downtime.
NFR-02.2	Scheduled maintenance activities shall be communicated to business stakeholders in advance.

NFR-03: Security

Requirement ID	Non-Functional Requirement
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NFR-03.1	The system shall enforce role-based access control (RBAC) for all users.
NFR-03.2	Sensitive business data shall be protected during storage and transmission.
NFR-03.3	The platform shall maintain audit logs for all critical business activities.
NFR-03.4	API access shall require secure authentication and authorization.

NFR-04: Reliability

Requirement ID	Non-Functional Requirement
NFR-04.1	The system shall maintain data consistency across merchant onboarding, payment, and settlement processes.
NFR-04.2	The platform shall record processing failures for operational investigation and recovery.

NFR-05: Scalability

Requirement ID	Non-Functional Requirement
NFR-05.1	The platform shall support increasing merchant volumes without significant system redesign.
NFR-05.2	The solution shall accommodate future business expansion and additional payment channels.

NFR-06: Compliance

Requirement ID	Non-Functional Requirement
NFR-06.1	The platform shall support regulatory compliance requirements applicable to merchant onboarding and payment processing.
NFR-06.2	The system shall retain audit records in accordance with organizational compliance policies.

NFR-07: Usability

Requirement ID	Non-Functional Requirement
NFR-07.1	The platform shall provide a consistent and intuitive user interface for business users.
NFR-07.2	The system shall present validation messages and error notifications in clear business

	language.
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NFR-08: Maintainability

Requirement ID	Non-Functional Requirement
NFR-08.1	The solution shall support future enhancements with minimal impact on existing functionality.
NFR-08.2	Configuration changes shall be manageable without major application redesign wherever applicable.

SECTION 12 — Business Rules

Business Rules

The following business rules define the operational policies that govern merchant onboarding, payment processing, fraud management, settlement processing, and compliance activities within the proposed platform.

Merchant Onboarding

Rule ID	Business Rule
BR-01	The system shall assign a unique Merchant ID upon successful merchant registration.
BR-02	A merchant profile shall remain in Pending Review status until KYC verification is completed.
BR-03	Merchants with incomplete mandatory information shall not proceed to KYC verification.

KYC & Compliance

Rule ID	Business Rule
BR-04	The system shall verify merchant KYC details through authorized external verification services.
BR-05	Merchant status shall be updated as Approved, Rejected, or Pending Review based on the verification outcome.

BR-06	Payment terminal activation shall not be permitted unless the merchant's KYC status is Approved.
BR-07	All KYC verification activities shall be recorded for audit and compliance purposes.

Payment Processing

Rule ID	Business Rule
BR-08	Only active merchant terminals shall be allowed to process payment transactions.
BR-09	Every successful payment transaction shall be assigned a unique Transaction ID.
BR-10	The system shall record transaction status as Success, Failed, or Suspicious Hold.

Fraud Management

Rule ID	Business Rule
BR-11	Transactions identified as high risk shall be placed under review before settlement.
BR-12	Fraud analysts shall be authorized to release or reject transactions under review.
BR-13	All fraud review activities shall be recorded in the audit log.

Settlement Management

Rule ID	Business Rule
BR-14	Only successful transactions that have passed fraud review, where applicable, shall be eligible for settlement.
BR-15	Transactions under fraud review shall be excluded from merchant settlement until a final decision is recorded.
BR-16	Each settlement cycle shall generate a unique settlement reference for reconciliation purposes.

Reporting & Audit

Rule ID	Business Rule
BR-17	The system shall maintain complete audit logs for merchant onboarding, payment, fraud, and settlement activities.
BR-18	Business reports shall display only authorized data based on user roles and permissions.

BR-19	Operational dashboards shall present the latest available transaction and settlement information.
BR-20	Audit records shall be retained in accordance with organizational compliance policies.

SECTION 13 — System Architecture

System Architecture

The proposed Merchant Onboarding, Payment Processing & Risk Management Platform follows a modular architecture that integrates merchant onboarding, regulatory verification, payment processing, fraud monitoring, settlement processing, and business reporting into a centralized platform.

The solution enables seamless communication between internal platform components and external verification services through secure REST APIs, ensuring automated workflows, regulatory compliance, and executive visibility.

High-Level System Architecture



Architecture Components

Component	Description
Merchant Portal	Allows merchants to register, submit business details, and track onboarding status.
Merchant Onboarding Service	Manages merchant registration, profile creation, and onboarding workflow.
KYC Verification API	Integrates with external verification services to validate merchant information.
Merchant Database	Stores merchant profiles, onboarding status, and compliance information.
Terminal Management Service	Provisions and manages merchant payment terminals after successful verification.
Payment Processing Engine	Processes payment transactions initiated through active merchant terminals.
Fraud Detection Engine	Evaluates transactions for potential fraud before settlement.
Transaction Database	Stores payment transaction details and processing status.
Settlement Engine	Calculates merchant settlements and manages payout processing.
Reporting & Analytics	Generates operational reports and business insights using SQL analytics and Tableau dashboards.

External Integrations

Integration	Purpose
KYC Verification Service	Merchant identity and compliance verification.
Tax Verification Service	Validation of business registration and tax information.
Payment Gateway	Secure payment transaction processing.
Banking Network	Merchant settlement and fund transfers.

Business Benefits

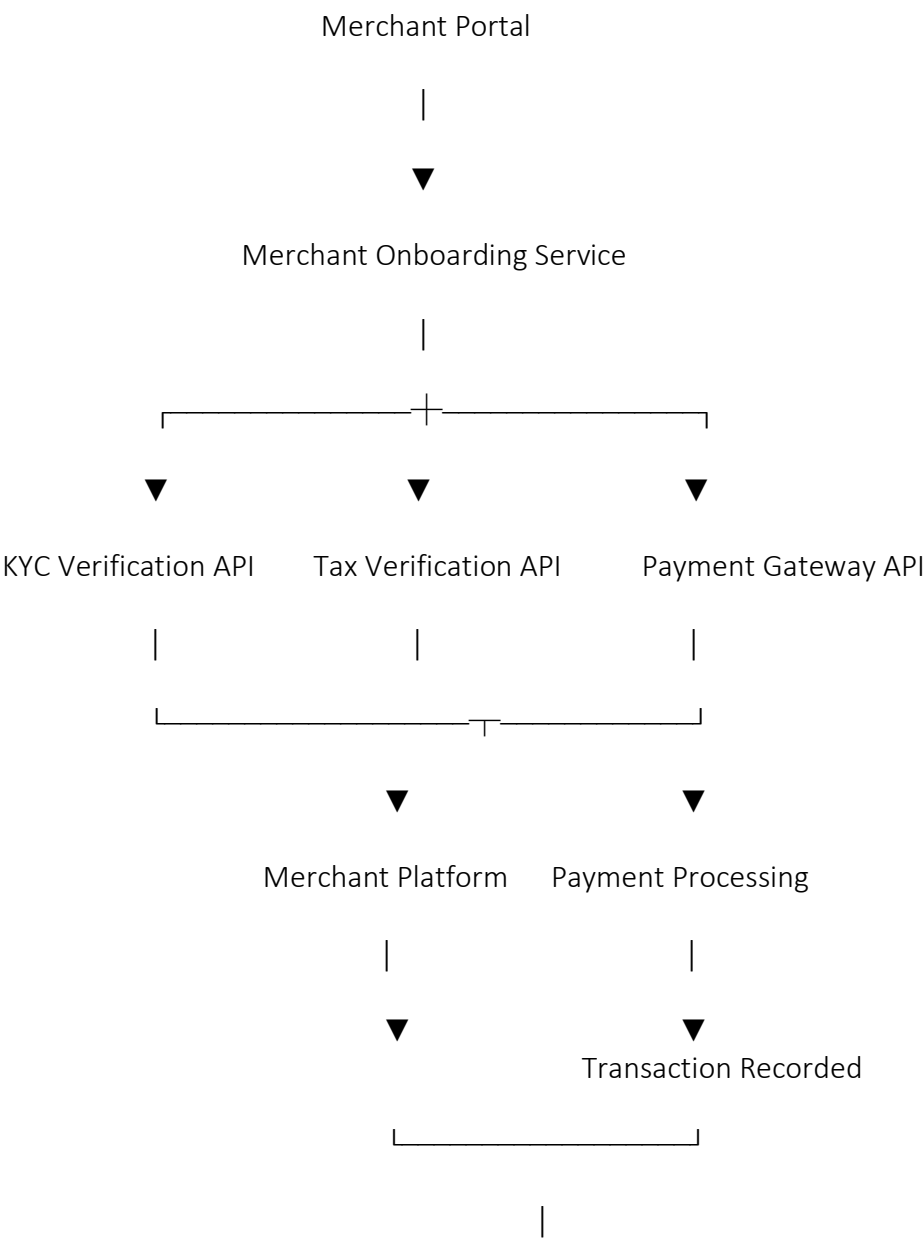
- Centralized merchant lifecycle management.
- Faster onboarding through API-driven verification.
- Secure payment processing with integrated fraud monitoring.
- Automated settlement processing and reconciliation.
- Improved executive visibility through business intelligence dashboards.
- Scalable architecture supporting future business growth.

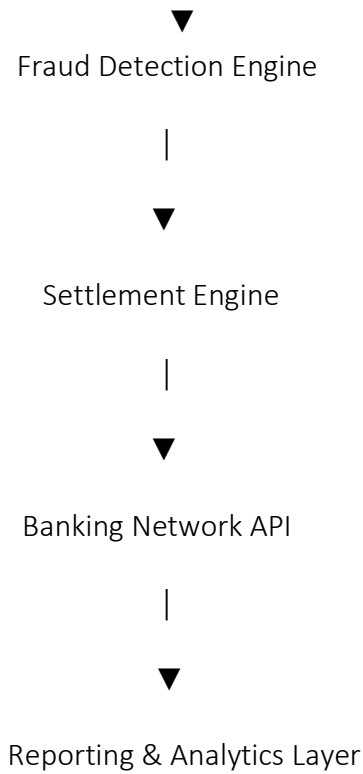
SECTION 14 — API Integration Overview

The proposed platform integrates with multiple external and internal systems through secure REST APIs to automate merchant onboarding, regulatory verification, payment processing, fraud assessment, and settlement operations.

These integrations eliminate manual verification activities, improve processing speed, and ensure seamless data exchange between business applications while maintaining regulatory compliance and operational reliability.

API Integration Architecture





API Integrations

API	Purpose	Business Value
Merchant Registration API	Creates and manages merchant profiles.	Standardizes merchant onboarding.
KYC Verification API	Validates merchant identity and compliance details.	Reduces manual verification effort and improves compliance.
Tax Verification API	Validates business registration and tax identifiers.	Improves regulatory accuracy.
Payment Gateway API	Processes customer payment transactions.	Enables secure payment processing.
Fraud Assessment API	Evaluates transaction risk before settlement.	Reduces fraud exposure.
Settlement API	Calculates merchant payouts and settlement status.	Automates settlement operations.
Notification API	Sends merchant onboarding and transaction notifications.	Improves merchant communication.

Integration Flow

1. Merchant submits registration details.
2. Merchant Registration API creates a merchant profile.

3. KYC Verification API validates business identity.
4. Tax Verification API verifies regulatory information.
5. Approved merchants receive payment terminal activation.
6. Payment Gateway API processes the transaction.
7. Transaction is recorded with a unique Transaction ID.
8. Fraud Assessment API evaluates transaction risk.
9. Low-risk transactions proceed to settlement; high-risk transactions are placed on hold for review.
10. Settlement API calculates payouts for eligible transactions.

Integration Benefits

- Automated merchant verification.
- Faster onboarding and activation.
- Reduced manual processing.
- Secure transaction processing.
- Improved fraud monitoring.
- Faster settlement execution.
- Centralized performance visibility.
- Scalable integration framework for future services.

SECTION 15 — API Contracts

The platform integrates with external verification services, payment gateways, and banking systems through REST APIs. The following API contracts define the business purpose, request structure, expected responses, validation rules, and error scenarios for key platform integrations.

API-01: Merchant Registration API

Business Purpose

Creates a new merchant profile and initiates the onboarding workflow.

Endpoint

POST /api/v1/merchants

Request

```
{  
  "businessName": "ABC Retail Pvt Ltd",  
  "businessType": "Retail",  
  "gstIn": "27ABCDE1234F1Z5",  
  "pan": "ABCDE1234F",
```

```
"contactName": "Rahul Sharma",  
"email": "rahul@abcretail.com",  
"mobile": "9876543210"  
}
```

Success Response (201 Created)

```
{  
  "merchantId": "M100001",  
  "status": "Pending Review",  
  "message": "Merchant registration submitted successfully."  
}
```

Business Validations

- Business Name is mandatory.
- GSTIN and PAN must be in a valid format.
- Email and mobile number must be unique.
- Merchant ID is generated automatically.

API-02: KYC Verification API

Business Purpose

Verifies merchant identity using an external compliance service.

Endpoint

POST /api/v1/kyc/verify

Request

```
{  
  "merchantId": "M100001",  
  "gstIn": "27ABCDE1234F1Z5",  
  "pan": "ABCDE1234F"  
}
```

Success Response

```
{  
  "merchantId": "M100001",  
  "kycStatus": "Approved",  
  "verificationDate": "2026-08-01"  
}
```

Failure Response — HTTP 400

```
{
  "merchantId": "M100001",
  "kycStatus": "Rejected",
  "reason": "GSTIN validation failed."
}
```

Business Validations

- Merchant must exist.
- GSTIN and PAN must match external verification records.
- Merchant status is updated automatically after verification.

API-03: Payment Processing API

Business Purpose

Processes payment transactions initiated by approved merchant terminals.

Endpoint

POST /api/v1/payments

Request

```
{
  "merchantId": "M100001",
  "terminalId": "TERM1001",
  "transactionAmount": 2500.00,
  "currency": "INR"
}
```

Success Response

```
{
  "transactionId": "TXN500001", "status": "Success",
  "message": "Payment processed successfully."
}
```

Error Response — HTTP 409

```
{
  "status": "Rejected",
  "errorCode": "PAY-409",
  "message": "Duplicate transaction request"
}
```

Business Validations

- Merchant must be active.
- Terminal must be active.
- Transaction amount must be greater than zero.
- Payment transactions shall undergo fraud risk assessment after transaction processing and before settlement eligibility is determined.

API-04: Settlement API

Business Purpose

Calculates merchant settlement amounts for eligible transactions.

Endpoint

POST /api/v1/settlements

Request

```
{
  "merchantId": "M100001",
  "settlementDate": "2026-08-02"
}
```

Success Response

```
{
  "settlementId": "SET200001",
  "status": "Processed",
  "netSettlementAmount": 245000.00
}
```

Business Validations

- Only successful transactions are eligible.
- Transactions under fraud review are excluded.
- Settlement record is generated for every processing cycle.
- Each settlement record shall reference the eligible transactions included in the settlement cycle.

Standard HTTP Status Codes

Status Code	Meaning
200	Request processed successfully
201	Resource created successfully
400	Invalid request or missing mandatory information
401	Unauthorized access

404	Requested resource not found
409	Duplicate merchant or conflicting request
500	Internal server error

API Security

- REST APIs secured using HTTPS.
- Authentication and authorization required for all endpoints.
- Sensitive business information protected during transmission.
- API requests and responses logged for audit and monitoring.

SECTION 16 — User Stories & Acceptance Criteria

The following user stories represent the key functional capabilities required to support merchant onboarding, compliance verification, payment processing, fraud management, and settlement operations. Each story includes business-oriented acceptance criteria using the Gherkin format.

Epic 1 – Merchant Onboarding & KYC

US-01: Merchant Registration

As a Merchant

I want to register my business on the platform

So that I can start accepting digital payments.

Acceptance Criteria

Given a merchant enters all mandatory business details

When the registration form is submitted

Then the system shall create a merchant profile with a status of Pending Review.

US-02: KYC Verification

As a Compliance Officer

I want to verify merchant KYC details

So that only eligible merchants are approved.

Acceptance Criteria

Given a merchant profile is awaiting verification

When the KYC verification API returns a successful response

Then the merchant status shall be updated to Approved.

US-03: Merchant Rejection

As a Compliance Officer
I want to reject merchants with invalid KYC information
So that regulatory compliance is maintained.

Acceptance Criteria

Given the KYC verification fails
When the validation response is received
Then the merchant status shall be updated to Rejected and the reason shall be recorded.

Epic 2 – Terminal & Payment Processing

US-04: Terminal Provisioning

As an Operations Executive
I want to activate a payment terminal only for approved merchants
So that payment processing can begin securely.

Acceptance Criteria

Given a merchant's KYC status is Approved
When terminal provisioning is initiated
Then the system shall generate terminal credentials and activate the terminal.

US-05: Payment Processing

As a Merchant
I want to process customer payments
So that I can receive payments digitally.

Acceptance Criteria

Given the merchant and terminal are active
When a payment request is received
Then the transaction shall be processed and a unique Transaction ID shall be generated.

Epic 3 – Fraud Management

US-06: Fraud Detection

As a Fraud Analyst
I want to identify high-risk transactions before settlement
So that financial losses are minimized.

Acceptance Criteria

Given a payment transaction has been processed and recorded
When the fraud assessment exceeds the defined risk threshold
Then the transaction shall be marked as Suspicious Hold.

US-07: Fraud Review

As a Fraud Analyst
I want to review held transactions
So that genuine transactions can proceed to settlement.

Acceptance Criteria

Given a transaction is under review
When the analyst approves the transaction
Then the transaction status shall be updated to Released and the transaction shall be eligible for the next settlement cycle.

Epic 4 – Settlement

US-08: Settlement Processing

As a Finance Executive
I want to calculate merchant settlements
So that merchants receive timely payouts.

Acceptance Criteria

Given eligible transactions are available
When the settlement process is executed
Then the system shall generate settlement records and payout references and create settlement details linking each eligible transaction to the settlement.

US-09: Settlement Exclusion

As a Finance Executive
I want to exclude transactions under fraud review
So that incorrect payouts are prevented.

Acceptance Criteria

Given a transaction has the status Suspicious Hold
When settlement is calculated
Then transaction shall not be linked to the settlement through a Settlement Detail record.

Epic 5 – Reporting

US-10: Operational Dashboard

As an Operations Manager
I want to view merchant, transaction, fraud, and settlement metrics
So that I can monitor platform performance and make informed decisions.

Acceptance Criteria

Given operational data is available
When the dashboard is accessed
Then the latest KPIs shall be displayed.

SECTION 17 — Low-Fidelity Wireframes

Wireframe Overview

The proposed platform provides dedicated interfaces for merchant onboarding, compliance verification, payment monitoring, fraud investigation, settlement processing, and operational reporting. These wireframes represent the primary workflows used by business and operations teams.

Screen 1 — Merchant Registration

VeriPay

Merchant Onboarding

Compliance Officer

Compliance > Verification Queue

Merchant Registration wireframe

Reviewing onboarding documentation for merchant account approval.

Internal Reviewer #62

84% IDB remaining

Merchant Name

ABC Retail Pvt Ltd

Merchant ID

M100001

KYC Status

PENDING REVIEW

Risk Rating

LOW

Submitted Documents / Verification

4 Documents Total

GSTIN Document

VERIFIED

View

PAN Card

VERIFIED

View

Business Registration Document

SUBMITTED

View

Address Proof (Utility Bill)

SUBMITTED

View

Verification Summary

GSTIN Status

VERIFIED

PAN Status

VERIFIED

Overall Status

PENDING REVIEW

SYSTEM ALERT LOG

Document OCR Match

Confidence score 98.5%

Watchlist Screening

No matches found in OFAC databases.

Domain Verification

Website content pending manual audit.

COMPLIANCE TIP

Ensure that the Address Proof matches the Registered Office address provided in the Business Registration Document.

Compliance Officer Actions

REVIEW NOTES

INTERNAL USE ONLY

Add observations about business legitimacy, document clarity, or discrepancy details...

Notes: Rejection requires a specific reason and will trigger an automated notification to the merchant with guidance for re-submission.

Approve KYC

Reject

Request Additional Documents

Save Progress

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PCI-DSS LEVEL 1 CERTIFIED SECURE PAYMENT ENVIRONMENT

Purpose

Allows merchants to register and submit business information for onboarding.

Key Components

- Business Name
- Business Type
- GSTIN
- PAN
- Contact Person
- Email Address
- Mobile Number
- Business Address
- Upload Business Documents
- **Submit Application** button

Screen 2 — KYC Verification Dashboard

VeriPay

Merchant Onboarding

Compliance Officer

Compliance > Verification Queue

KYC Verification Dashboard

OFFICER ASSIGNED: Internal Reviewer 002

Reviewing onboarding documentation for merchant account approval.

SLA: 12h remaining

MERCHANT NAME

ABC Retail Pvt Ltd

MERCHANT ID

PM00001

KYC STATUS

Pending Review

RISK INDICATOR

Low

Submitted Documents / Verification

6 Documents Total

Document Icon

GSTIN Document

Verified

View

Document Icon

PAN Card

Verified

View

Document Icon

Business Registration Document

Submitted

View

Document Icon

Address Proof (Utility Bill)

Submitted

View

Verification Summary

GSTIN Verification Status

Verified

PAN Verification Status

Verified

Overall KYC Status

Pending Review

SYSTEM AUDIT

Document OCR Match

Confidence score: 98.5%

Watchlist Screening

No matches found in AML databases.

Domain Verification

Website content passing manual audit.

Compliance Tip

Ensure that the Address Proof matches the Registered Office address provided in the Business Registration Document.

Compliance Officer Actions

REVIEW NOTE

INTERNAL USE ONLY

Add observations about business legitimacy, document clarity, or discrepancy details...

Notes: Rejection requires a specific reason and will trigger an automated notification to the merchant with guidance for re-submission.

Approve KYC

Reject

Request Additional Documents

Save Progress

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PCI-DSS LEVEL 1 CERTIFIED SECURE PAYMENT ENVIRONMENT

Purpose

Allows Compliance Officers to review and approve merchant verification requests.

Key Components

- Merchant ID
- Merchant Name
- Submitted Documents
- GSTIN Verification Status
- PAN Verification Status
- KYC Status
- Risk Indicator
- **Approve**
- **Reject**
- **Request Additional Documents**

Screen 3 — Merchant Dashboard



Purpose

Provides merchants with onboarding progress and operational information.

Key Components

- Merchant Profile
- KYC Status
- Terminal Status
- Recent Transactions
- Settlement Status
- Notifications
- Support Requests

Screen 4 — Transaction Monitoring Dashboard



Purpose

Allows Operations Teams to monitor payment transactions.

Key Components

- Transaction ID
- Merchant Name
- Transaction Amount
- Payment Status
- Processing Time
- Payment Method
- Search & Filters

Screen 5 — Fraud Review Dashboard

VeriPay

Merchant Onboarding

Compliance Officer

Back to Queue

Fraud Control

Case #TTSN000001

Transaction Investigation

Simplex Hold

Assign to Me

1

Manual review required due to high-risk velocity checks and atypical merchant-category spending patterns detected within a 5-minute window.

DISPICUOUS TRANSACTION DETAIL

TRANSACTION INFO

TRANSACTION ID

123456789

AMOUNT

€2,500.00

ACQ ON DATE

18 Jul 2020, 11:42 AM

PAYMENT METHOD

WALA DIGIT **** 4382

MERCHANT ENTITY

REGISTERED NAME

ABC Retail Pvt Ltd

MERCHANT ID

802-832918

View Profile

ORIGINAL LOCATION

Mumbai, MH (IP: 103.21.11.85)

ORIGIN IP RANGE

Mobile App (CG: 1/3)

STATUS

Level 3 (Verified)

RISK INFORMATION

88

RISK SCORE INDEX

Velocity Check

Failed (4/50s)

IP Reputation

Medium Risk

Global Watchlist

No Match

FRAUD TRIGGER

SYSTEM FLAG/CATEGORY

"Unusual transaction pattern"

REASONING (1/200)

Transaction amount exceeds merchant average by 80%.

Multiple authorization attempts from same IP range in 60s.

AUDIT HISTORY

Transaction Held - 18 Jul, 11:42 AM
By: System Sentinel
Velocity Check succeeded. Rulebook ID: 002-100

Pinged for Manual Review - 18 Jul, 11:42 AM
By: Risk Engine v2.0

Authorization Initiated - 18 Jul, 11:42 AM
By: Visa Network Gateway

VIEW FULL TRACE

INVESTIGATION TOOLKIT

First Order

Chargebacks

IP History

Global Search

ANALYZING IP

Check for "True Value" auth attempts prior to this transaction.
Fraudsters often use them to validate stolen card credentials.

INVESTIGATION NOTES & ACTION

INVESTIGATION SUMMARY

Detail your findings... e.g., "Called merchant for verification", "Customer confirmed order", etc.

Save

Release Transaction

Reject Transaction

Escalate Case

COMPLIANCE STATUS

Good Standing

Active

Account Age

3.2 Years

Outstanding

0

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PCI-DSS LEVEL 1 CERTIFIED SECURE PAYMENT ENVIRONMENT

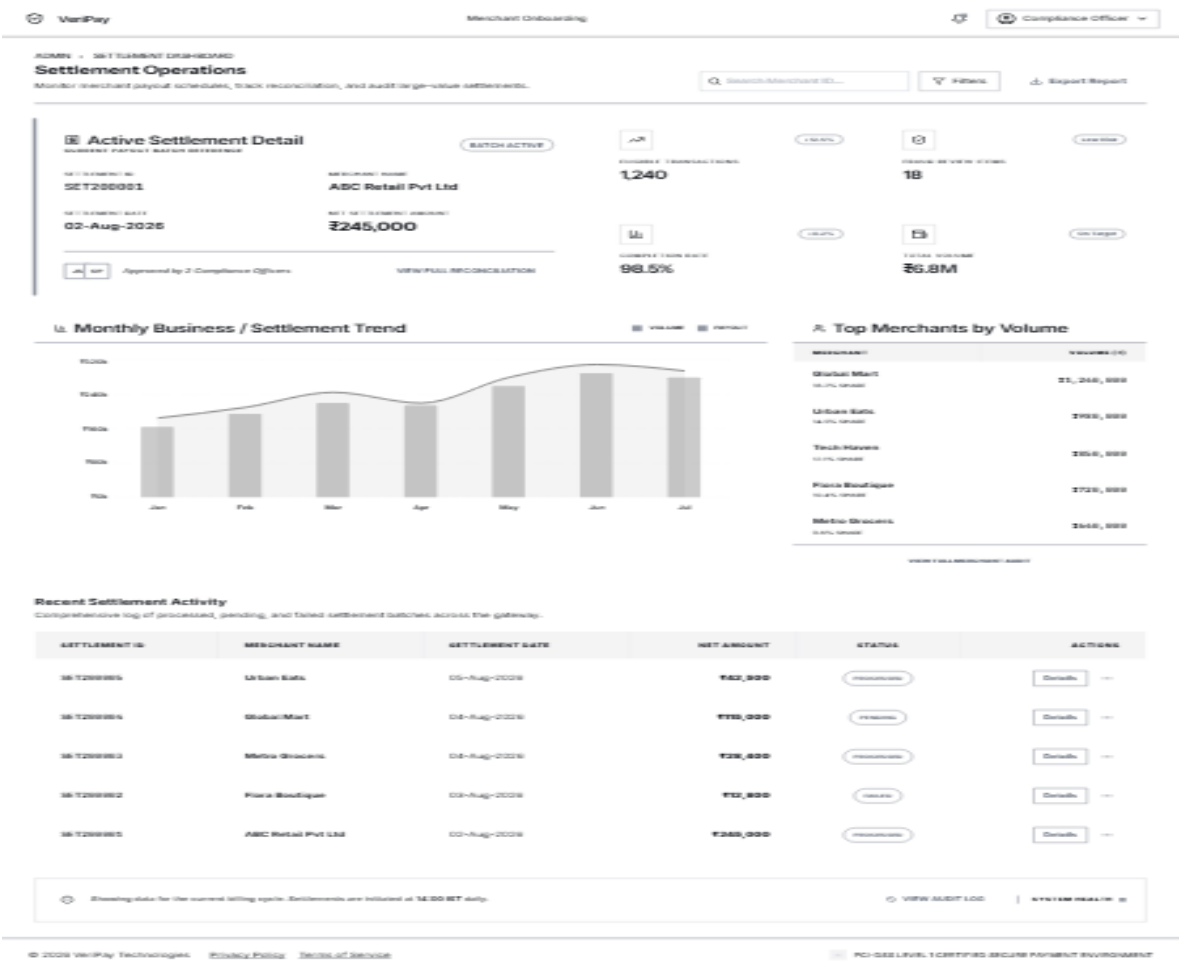
Purpose

Allows Fraud Analysts to investigate suspicious transactions.

Key Components

- Transaction ID
- Merchant Name
- Risk Score
- Fraud Trigger
- Investigation Notes
- **Release Transaction**
- **Reject Transaction**
- Audit History

Screen 6 — Settlement Dashboard



Purpose

Allows Finance Teams to monitor merchant payouts.

Key Components

- Settlement ID
- Merchant Name
- Settlement Date
- Net Settlement Amount
- Fraud Cases Under Review
- Settlement Completion Rate
- Top Merchants by Transaction Volume
- Monthly Business Trends

Navigation Flow

Merchant Registration

|



KYC Verification

|



Merchant Dashboard

|



Transaction Monitoring

|



Fraud Review

|



Settlement Dashboard

|

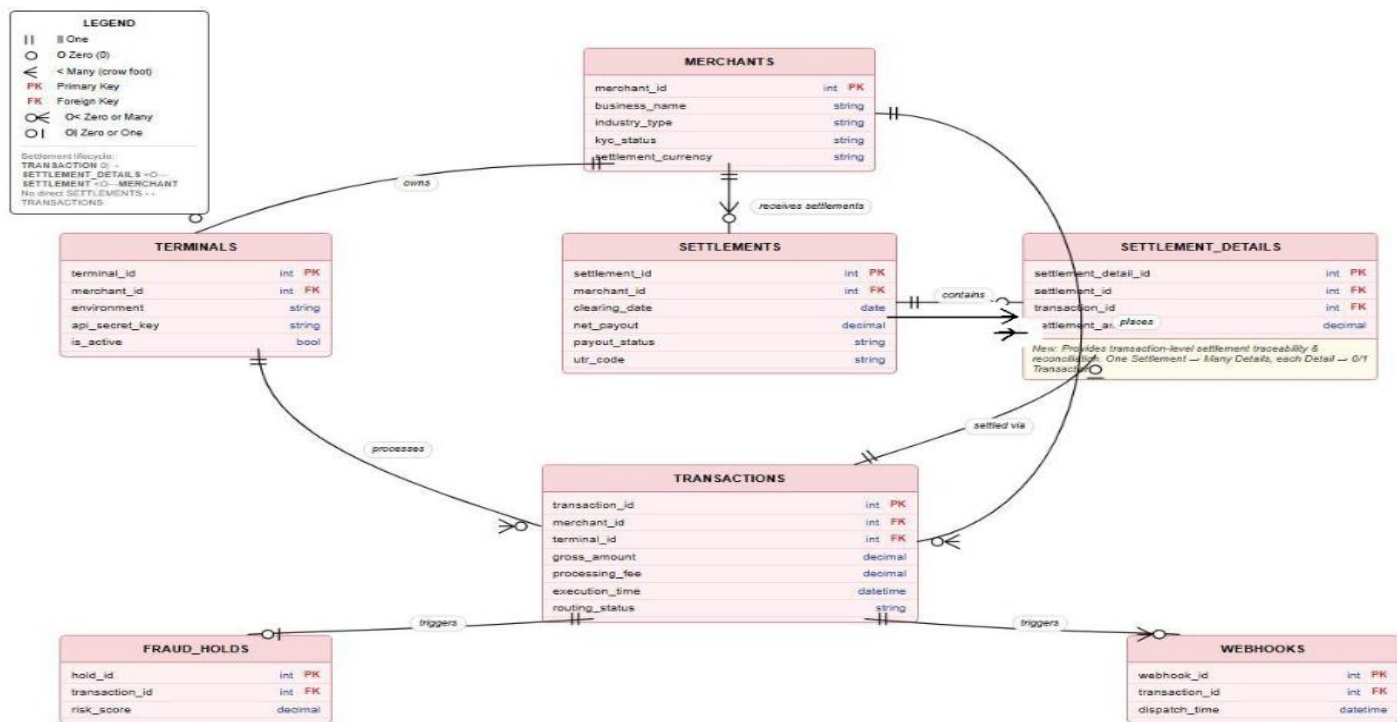


Executive Dashboard

SECTION 18 — Entity Relationship Diagram

The Entity Relationship Diagram (ERD) represents the logical data model for the Merchant Onboarding, Payment Processing & Risk Management Platform. It illustrates how merchant information, payment terminals, transactions, fraud investigations, webhook notifications, and settlement records are related to support the end-to-end payment lifecycle.

The data model has been designed to maintain data integrity, reduce redundancy, and support reporting, reconciliation, fraud monitoring, and settlement processing.



Entity Relationships

Parent Entity	Child Entity	Relationship
Merchants	Terminals	One Merchant → Many Terminals
Merchants	Transactions	One Merchant → Many Transactions
Merchants	Settlements	One Merchant → Many Settlement Records
Terminals	Transactions	One Terminal → Many Transactions
Transactions	Fraud Holds	One Transaction → Zero or One Fraud Hold
Transactions	Webhooks	One Transaction → Many Webhook Events

Data Model Highlights

- Merchant records serve as the master entity across the platform.
- Each approved merchant can have multiple payment terminals.
- Every transaction is linked to both a merchant and a terminal.
- High-risk transactions are tracked separately through Fraud Holds.
- Webhook events maintain communication history with external systems.
- Settlement records maintain payout history for reconciliation and financial reporting.

Business Benefits

- Supports complete merchant lifecycle management.
- Ensures referential integrity across operational data.
- Simplifies reporting and SQL analytics.
- Enables efficient fraud investigation and settlement reconciliation.
- Provides a scalable foundation for future platform enhancements.

SECTION 19 — SQL Database Schema Overview

The proposed platform is supported by a normalized relational database designed to manage merchant onboarding, payment processing, fraud monitoring, settlement management, and operational reporting.

The database structure ensures data integrity, minimizes redundancy, and provides a scalable foundation for transaction processing, business analytics, and regulatory reporting.

The supporting database implementation is developed using PostgreSQL, with referential integrity enforced through primary keys, foreign keys, and validation constraints.

Database Design Principles

The database has been designed to:

- Reduce data redundancy through normalized tables.
- Support secure transaction processing.
- Enable efficient reporting and business analytics.
- Provide scalability for future merchant and transaction growth.

Database Tables

Table	Business Purpose
Merchants	Stores merchant profile information, KYC status, and onboarding details.
Terminals	Maintains payment terminal configuration and activation status for each merchant.
Transactions	Records all payment transactions processed through merchant terminals.
Fraud Holds	Stores high-risk transactions pending fraud investigation.
Webhooks	Tracks outbound notification events exchanged with external systems.
Settlements	Maintains merchant payout records and settlement history.
Settlement_Detail	Maintains transaction-level details for each settlement, supporting settlement reconciliation and traceability.

Primary & Foreign Key Relationships

Table	Primary Key	Foreign Keys
Merchants	Merchant_ID	—
Terminals	Terminal_ID	Merchant_ID
Transactions	Transaction_ID	Merchant_ID, Terminal_ID
Fraud Holds	Hold_ID	Transaction_ID
Webhooks	Webhook_ID	Transaction_ID
Settlements	Settlement_ID	Merchant_ID
Settlement_Detail	Settlement_Detail_ID	Settlement_ID, Transaction_ID

Data Integrity Rules

The database enforces several integrity rules to ensure consistent and reliable business operations.

- Every merchant is assigned a unique Merchant ID.
- Payment terminals can only be associated with existing merchants.
- Transactions can only be processed through registered terminals.
- Fraud Hold records can only reference valid transactions.
- Settlement records can only be generated for registered merchants.
- Referential integrity is enforced using foreign key relationships.
- Mandatory business fields are validated using database constraints.
- Each settlement detail record shall reference a valid settlement and transaction.

Business Benefits

- Reliable and consistent business data.
- Improved transaction traceability.
- Faster settlement reconciliation.
- Simplified SQL reporting and analytics.
- Better regulatory compliance through structured data management.
- Scalable database architecture supporting business growth.

SECTION 20 — Sprint Backlog

The proposed solution is structured using an Agile Scrum approach. The product backlog has been prioritized into four implementation sprints, enabling incremental delivery of business capabilities while allowing continuous stakeholder feedback throughout the development lifecycle.

Sprint 1 — Merchant Onboarding & KYC

Sprint Goal

Establish the merchant onboarding process and automate compliance verification.

User Story	Priority
Merchant Registration	High
Merchant Profile Management	High
KYC Verification	High
Merchant Approval / Rejection	High
Notification on KYC Status	Medium

Sprint Deliverables

- Merchant Registration Module
- KYC Verification Workflow
- Merchant Status Management

- Notification Service
- Maintain referential integrity across all business entities.

Sprint 2 — Terminal & Payment Processing

Sprint Goal

Enable merchants to process digital payments securely.

User Story	Priority
Terminal Provisioning	High
Terminal Activation	High
Payment Processing	High
Transaction History	Medium
Payment Status Tracking	Medium

Sprint Deliverables

- Terminal Management
- Payment Processing Engine
- Transaction Management
- Payment Status Dashboard

Sprint 3 — Fraud Management & Settlement

Sprint Goal

Improve transaction security and automate settlement operations.

User Story	Priority
Fraud Detection	High
Fraud Review	High
Settlement Processing	High
Settlement Exclusion	High
Settlement History	Medium

Sprint Deliverables

- Fraud Detection Module
- Fraud Review Dashboard
- Settlement Engine
- Settlement Tracking

Sprint 4 — Reporting & Analytics

Sprint Goal

Provide business users with operational insights and reporting capabilities.

User Story	Priority
Executive Dashboard	High
Merchant Performance Dashboard	Medium
Transaction Analytics	High
Settlement Analytics	Medium
Operational Reports	Medium

Sprint Deliverables

- Executive Dashboard
- SQL Analytics
- Tableau Dashboard
- Business Reports

Product Backlog Prioritization

Priority	Features
High	Merchant Registration, KYC Verification, Payment Processing, Fraud Detection, Settlement Processing, Executive Dashboard
Medium	Notifications, Transaction History, Merchant Performance Reporting, Settlement Analytics
Low	Future integrations, advanced reporting enhancements, additional payment channels

Definition of Ready (DoR)

A user story is considered Ready when:

- Business requirements are approved.
- Acceptance criteria are defined.
- Dependencies are identified.
- Stakeholder review is completed.
- Story is estimated by the development team.

Definition of Done (DoD)

A user story is considered Done when:

- Development is completed.
- Unit testing is successful.
- Functional testing is completed.
- Acceptance criteria are satisfied.

- Business approval is received.
- Documentation is updated.
- Solution is ready for deployment following business approval.

SECTION 21 — SQL Analytics Overview

The platform leverages advanced SQL analytics to transform operational data into actionable business insights. The reporting layer enables stakeholders to monitor merchant onboarding, transaction performance, fraud activity, settlement operations, and overall platform health through structured analytical queries.

The SQL analytics suite is implemented using PostgreSQL and consists of 25 business-focused analytical queries designed to support operational reporting, management decision-making, and business performance analysis.

Analytics Categories

Category	Business Purpose
Revenue Analytics	Monitor revenue trends and platform growth
Merchant Analytics	Monitor merchant onboarding and merchant performance
Transaction Analytics	Measure transaction success, failures and processing performance
Fraud Analytics	Identify suspicious transactions requiring investigation
Settlement Analytics	Monitor settlement processing and payout status
Financial Analytics	Analyze processing fees and merchant revenue contribution

Key Business Insights Generated

The SQL reporting suite provides insights including:

- Monthly revenue trend
- Merchant onboarding status
- Transaction success rate
- Fraud investigation summary
- Settlement status
- Merchant revenue ranking
- Industry-wise transaction distribution

- Processing fee analysis

Advanced SQL Concepts Used

The analytical queries demonstrate practical use of advanced SQL concepts, including:

- Common Table Expressions (CTEs)
- Window Functions
- Aggregate Functions
- CASE Expressions
- JOIN Operations
- Subqueries
- Date & Time Functions
- Ranking Functions
- Conditional Aggregation
- Data Grouping and Filtering

Business Value

The SQL analytics layer enables business stakeholders to:

- Monitor operational performance.
- Identify revenue trends.
- Improve merchant onboarding decisions.
- Detect fraud patterns proactively.
- Track settlement efficiency.
- Support data-driven business decisions.

Technical Implementation

Supporting implementation artifacts are available in the GitHub repository, including;

- Database Schema (DDL)
- Sample Data (DML)
- Advanced Analytics Queries
- Query Documentation
- Tableau Dashboard
- Project README

Sample Business Questions Answered

The analytics suite answers questions such as:

- How is monthly revenue trending?
- Which merchants generate the highest transaction value?
- What is the transaction success rate?
- Which merchants generate the highest revenue?
- Which transactions require fraud investigation?
- How efficiently are settlements processed?
- How much processing fee is incurred by each merchant?

SECTION 22 — Tableau Dashboard Overview

Tableau Dashboard Overview

An interactive Tableau dashboard has been developed to provide business stakeholders with a centralized view of merchant onboarding, payment transactions, fraud monitoring, settlement operations, and overall platform performance.

The dashboard enables decision-makers to monitor key operational metrics, identify trends, and support data-driven business decisions through visual analytics.

Dashboard Objectives

The dashboard has been designed to:

- Monitor merchant onboarding performance.
- Track transaction and revenue trends.
- Analyze payment processing performance.
- Monitor fraud investigation activities.
- Track settlement status and payout performance.
- Provide executive-level operational visibility.

Dashboard Component	Business Purpose
Merchant Overview	Displays merchant onboarding and activation metrics.
Transaction Performance	Monitors transaction volume and processing status.
Revenue Trends	Tracks revenue growth across reporting periods.

Operational KPIs	Presents key business performance indicators for management.
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Key Metrics Displayed

The dashboard includes the following business KPIs:

- Total Merchants
- Active Merchants
- Total Transactions
- Transaction Success Rate
- Total Revenue
- Monthly Revenue Trend

Business Value

The Tableau dashboard enables stakeholders to:

- Monitor business performance using the latest available operational data.
- Identify operational bottlenecks.
- Improve decision-making using visual insights.
- Track fraud and settlement performance.
- Measure merchant onboarding efficiency.
- Support executive reporting through centralized dashboards.

Tools & Technologies

Tool	Purpose
PostgreSQL	Database Management
SQL	Business Insights
Tableau Public	Interactive Dashboard & Data Visualization
GitHub	Version Control & Project Repository

SECTION 23 — Risks & Mitigation

The successful implementation of the Merchant Onboarding, Payment Processing & Risk Management Platform depends on proactively identifying potential project and operational risks. The following risk register outlines key risks, their potential impact, and the corresponding mitigation strategies.

Risk Register

Risk ID	Risk Description	Impact	Mitigation Strategy
---------	------------------	--------	---------------------

R-01	Delays in external KYC verification services	High	Implement retry mechanisms and manual review procedures for verification failures.
R-02	Incomplete or incorrect merchant information	Medium	Validate mandatory fields and enforce business validation rules during registration.
R-03	Payment gateway service interruptions	High	Enable monitoring, alerting, and fallback procedures for failed transactions.
R-04	False positive fraud detection	Medium	Provide manual fraud review before final transaction decisions.
R-05	Settlement processing delays	High	Automate settlement workflows and monitor pending settlement queues.
R-06	Unauthorized system access	High	Implement role-based access control (RBAC), authentication, and audit logging.
R-07	Data quality issues impacting reporting	Medium	Apply validation checks and periodic data quality reviews.
R-08	Increasing merchant and transaction volumes affecting performance	Medium	Design a scalable database and application architecture capable of supporting business growth.

Risk Prioritization Matrix

Priority	Risks
----------	-------

High	KYC Service Availability, Payment Gateway Failure, Settlement Delays, Unauthorized Access
Medium	Data Quality, Fraud False Positives, Merchant Data Errors, Platform Scalability
Low	Minor UI enhancements, non-critical reporting improvements

Risk Monitoring Approach

Project risks will be monitored throughout the implementation lifecycle by:

- Regular sprint reviews.
- Stakeholder status meetings.
- Operational dashboard monitoring.
- API integration health checks.
- Settlement reconciliation reviews.
- Security and audit log monitoring.
- Periodic risk assessment sessions.

Business Benefits of Risk Management

- Reduced operational disruptions.
- Improved regulatory compliance.
- Faster issue identification and resolution.
- Increased platform reliability.
- Better merchant experience.
- Improved business continuity.
- Enhanced stakeholder confidence.

SECTION 24 — Business Impact & Success KPIs

The proposed Merchant Onboarding, Payment Processing & Risk Management Platform delivers measurable improvements across merchant onboarding, compliance, payment processing, fraud management, settlement operations, and business reporting.

By automating key operational workflows and integrating business processes into a centralized platform, the solution improves operational efficiency, strengthens regulatory compliance, enhances customer experience, and provides greater visibility into business performance.

Expected Business Outcomes

Business Area	Expected Outcome
Merchant Onboarding	Faster onboarding through automated KYC verification and process automation.
Regulatory Compliance	Improved compliance through standardized verification and audit capabilities.
Payment Processing	More reliable and secure transaction processing.
Fraud Management	Earlier identification of high-risk transactions before settlement.
Settlement Operations	Faster and more accurate merchant payout processing.
Operational Efficiency	Reduced manual effort through process automation.
Business Reporting	Improved decision-making through centralized dashboards and SQL analytics.
Platform Scalability	Ability to support increasing merchant and transaction volumes with minimal operational impact.

Expected Business Value

The proposed solution is expected to deliver value by:

- Streamlining merchant onboarding through workflow automation.
- Strengthening compliance with standardized KYC verification.
- Improving payment processing reliability.
- Reducing fraud exposure through proactive monitoring.
- Accelerating settlement processing.
- Increasing operational efficiency by minimizing manual activities.
- Providing centralized operational reporting through SQL analytics and Tableau dashboards.
- Supporting scalable business growth with a modular platform architecture.

Project Deliverables Summary

The completed project includes:

- Business Requirements Document (BRD)
- Functional & Non-Functional Requirements
- Business Rules
- Stakeholder Matrix

- AS-IS & TO-BE Process Flows
- Gap Analysis
- System Architecture
- API Integration Overview
- API Contracts
- User Stories with Gherkin Acceptance Criteria
- Low-Fidelity Wireframes
- Entity Relationship Diagram (ERD)
- SQL Database Design
- PostgreSQL Implementation
- 25 Advanced SQL Analytics Queries
- Tableau Executive Dashboard
- Risk Register & Mitigation Plan
- GitHub Repository Documentation

Project Conclusion

This project demonstrates an end-to-end Business Analysis approach for designing a modular FinTech platform that supports merchant onboarding, payment processing, fraud management, settlement operations, and business intelligence reporting.

The solution combines business process analysis, requirements engineering, system design, API integration, SQL analytics, and data visualization into a single enterprise case study. It reflects how a Business Analyst collaborates with product, engineering, compliance, operations, and business stakeholders to translate business needs into practical, technology-enabled solutions.

